

Project Report: Enhanced Image Inpainting using Transformer-GAN Hybrid Models

Introduction:

The primary objective of this project is to significantly enhance image inpainting technology by effectively implementing and integrating Transformer architecture and Generative Adversarial Networks (GANs). For this project we have decided to select and work with CelebA dataset due to its excellent attributes that complement the objectives of this project. With 202,599 high-quality facial images, CelebA dataset is very diverse and complex and hence sufficient for providing diversity for enhancing robustness in deep learning models trained for image reconstruction tasks. CelebA's rich collection of facial attributes, expressions, and poses is particularly beneficial for learning detailed and intricate patterns necessary for realistic facial image inpainting. The dataset allows challenging cases, including irregular masks and large missing regions, that mimic real-world inpainting situations. Its prevalent use in past work proves it to be a valid benchmark, which has been continually verified by using metrics like PSNR and SSIM. Taking these strengths forward, this project aims at developing a Transformer-GAN hybrid model that not only enhances the image reconstruction quality but also sets a higher standard for image inpainting solutions.

Dataset and Environment Setup:

We established an organized environment in order to work with and preprocess CelebA efficiently. First, we replicated our compressed version of CelebA dataset (img_align_celeba.zip) from Google Drive and unzip it locally such that all files of images became readily accessible. To ease preprocessing and loading images, we utilized large Python libraries like PyTorch and torchvision for operations on the dataset and image transformations, pandas for metadata and annotation handling, and PIL for image-specific tasks like reading and conversion of formats. We then checked the integrity and completeness of the dataset after extraction through image quantity checks and ensured that metadata files were present. This diligent preparation made it compatible with our deep learning pipeline while keeping data quality in check essential to successful model training.

EDA:

We conducted a thorough Exploratory Data Analysis (EDA) to familiarize and prepare and know the CelebA dataset in-depth before training our hybrid model of Transformer-GAN. We first read and concatenated the dataset features and partition information utilizing pandas. We converted the attribute labels from the original (-1, 1) format to a more natural binary format (0, 1). We also divided the dataset into training (162,770 samples), validation (19,867 samples), and test sets (19,962 samples) as per the official splits of the dataset. To achieve deeper insight, we computed descriptive statistics for all attributes and revealed their distribution and prevalence in the data set. For instance, attributes like 'Attractive', 'Young', and 'Smiling' contained various distributions that were crucial in guiding our model's learning schedule. We also verified whether missing values

existed in each column of attributes to ascertain that the data set was whole and valid. Then, we've defined significant changes with torchvision like resizing, random horizontal flipping, and normalizing to prepare images efficiently for deep learning. We created a custom 'CelebADataset' class with proper error handling to only select existing and readable images, and data became more trustworthy. Using PyTorch's 'DataLoader', we prepared data batching efficiently for training, validation, and testing. Furthermore augmenting our data preparation, we introduced random masks to simulate realistic inpainting conditions in real-world scenarios by incorporating random square occlusions into images. This was achieved via the 'InpaintingCelebADataset' class, such that our model experiences realistic training conditions similar to those in real-world applications. Finally, we presented masked images, their corresponding masks, and original images alongside each other, verifying our masking strategy and dataset preparation. This thorough exploratory procedure ensured that our dataset was optimally structured and ready for effective Transformer-GAN hybrid model training.

Visualizations:

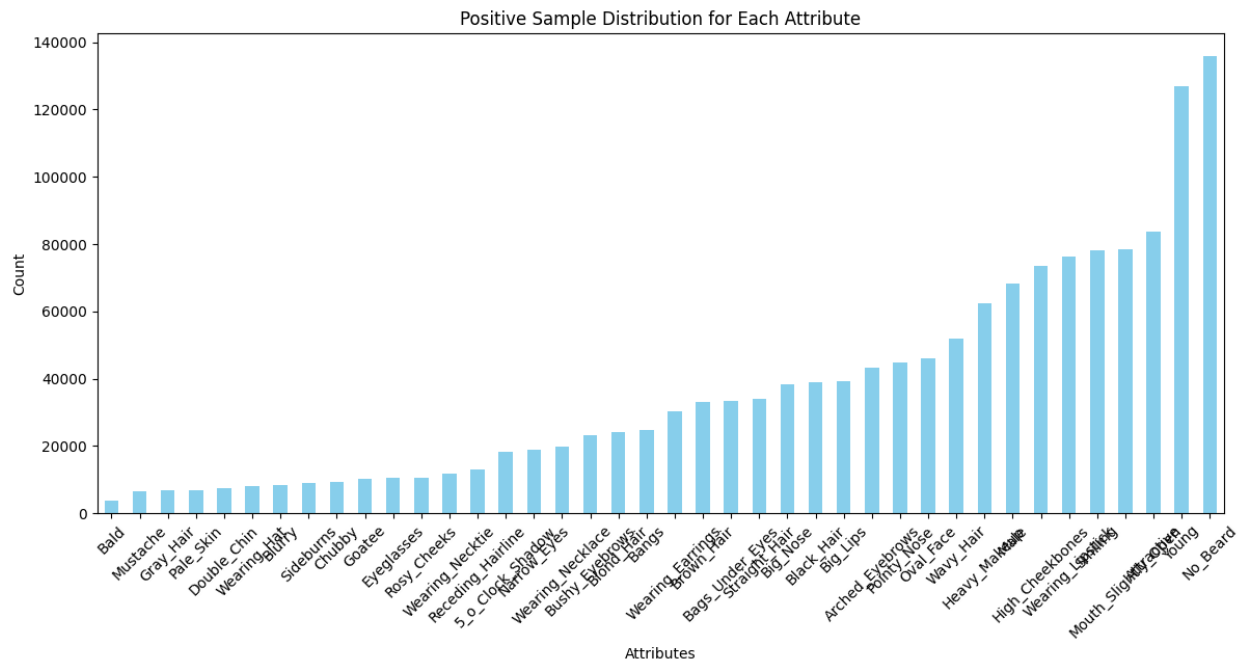


Figure1

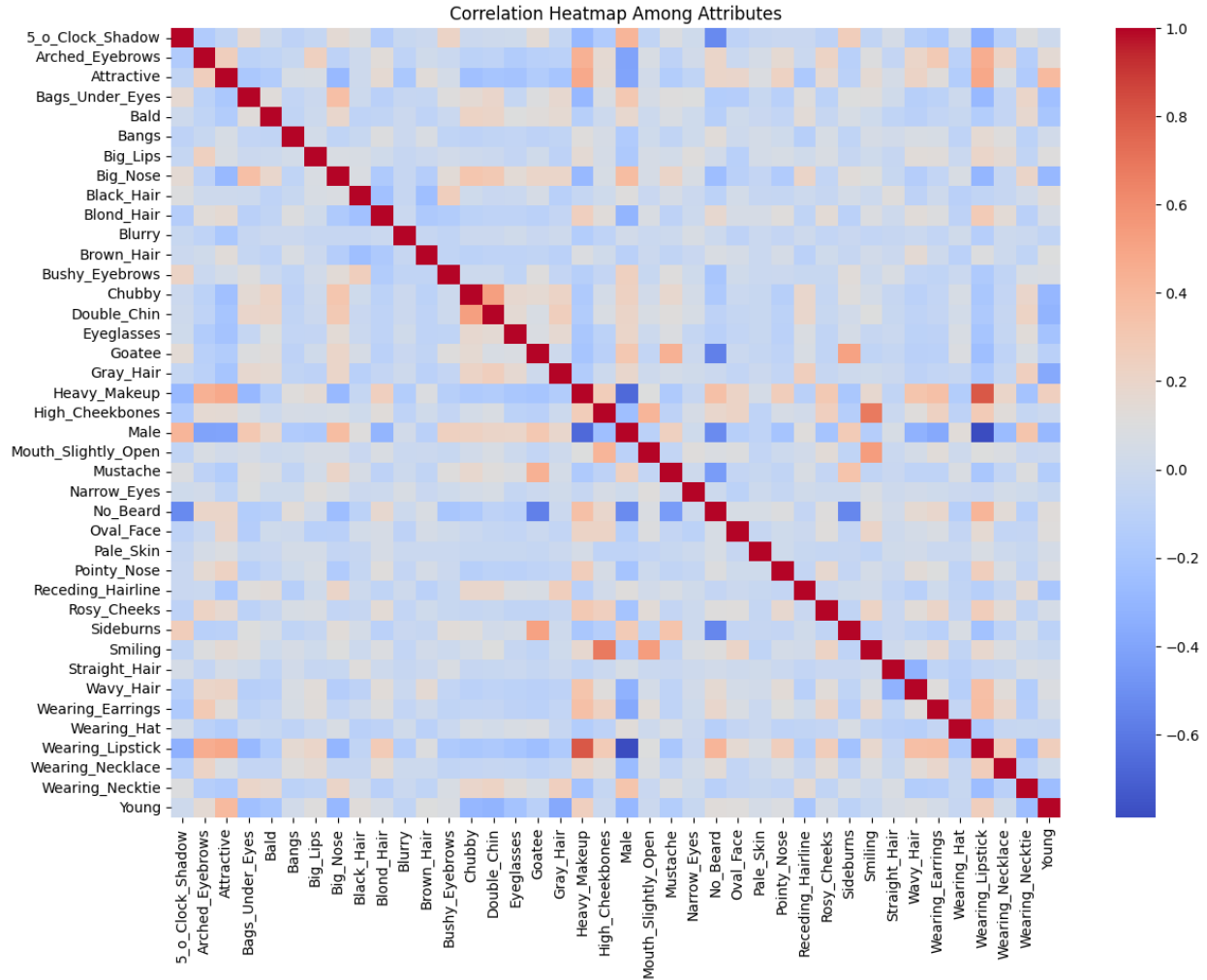


Figure2

We performed further exploratory visualizations to better understand the nature and interrelation within the CelebA dataset. Figure 1 depicts the number of positive samples per facial attribute in the training set, which clearly demonstrates the frequency and attribute imbalance between attributes. Surprisingly, attributes such as "No_Beard" and "Young" contained significantly more instances than attributes such as "Bald," "Mustache," and "Gray_Hair," affecting our perception of potential biases with model training.

In Figure 2, we present a correlation heatmap exploring correlations among the 40 features. By means of this visualization, we are able to identify strong correlations, both positive and negative, enabling further examination of dependencies and redundancies in features. For example, facial hair and gender features possessed stronger correlations, capturing patterns necessary for well-informed model design decisions. These graphical examinations were crucial in determining how to construct a good and robust Transformer-GAN hybrid model.

Methodology:

We developed a baseline model of Transformer-GAN hybrid model tailored for the CelebA dataset to solve the image inpainting task efficiently. The model architecture contains two basic components: generator and discriminator.

Generator Architecture:

The generator uses an encoder-decoder structure to effectively capture global information and synthesize high-level textures of images. In the encoder, we utilized successive convolutional layers with ReLU activation and batch normalization to iteratively extract and downsample required features. The decoder reverses the process with transpose convolutional layers to upsample and synthesize the entire image, utilizing Tanh activation to normalize the outputs within the range $[-1, 1]$.

Discriminator Architecture:

Our discriminator is designed to distinguish between model-generated images and real images and give realistic output. It comprises a number of convolutional layers with LeakyReLU activations and batch normalization for efficient feature extraction. After the final convolutional layer, a sigmoid function and adaptive average pooling are employed to produce a confidence score indicating the authenticity of an image.

Training Procedure:

Binary cross-entropy (BCE) loss during adversarial training guided the generator in producing images of realistic output, and one other L1 reconstruction loss as additional pixel fidelity for the reconstructed part. A 0.0002 learning rate Adam optimizer was chosen for good performance due to its apt choice, wherein separate optimizers for the discriminator and generator. Training loop included several epochs of iterative optimisation, which were alternately updated for discriminator and generator. We tracked and printed loss statistics per epoch to monitor convergence as well as stable training.

Results Visualization:

We quantified the visual quality of our inpainted results by examining generated images across several training epochs. The early outputs (Epoch 1) exhibited major reconstruction artifacts and softness in the masked regions. However, by Epoch 2, the detail and texture consistency improvements began to emerge. Additional training up to Epoch 3 and Epoch 4 resulted in progressively clearer and more natural-looking inpainting outputs, which indicates the model's growing capability to reconstruct missing image parts correctly.

After training, we stored the trained generator and discriminator models in final form to ensure reproducibility and simplicity of use in the future. At inference, we loaded the trained generator model to create inpainted images on unseen masked inputs. Inference visualization (Figure: "Inpainted Images (Inference)") confirms the generator's capability in face image restoration, providing improved coherence and realism in reconstruction regions. Apart from that, we also performed a quantitative performance evaluation on the test data. We used the common metrics of Mean Absolute Error (L1 Loss), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). We achieved a mean L1 Loss of 0.0377, representing small reconstruction error. Further, our model achieved a mean PSNR of 23.75 dB, indicating quality restoration, and an SSIM value of 0.7674, which represents high structural fidelity of produced and original images.

Training Images:

Inpainted Images at Epoch 1



Inpainted Images at Epoch 2



Inpainted Images at Epoch 3



Inpainted Images at Epoch 4



Inpainted Images at Epoch 5



Inpainted Images (Inference)



Visualization Graphs & analysis:

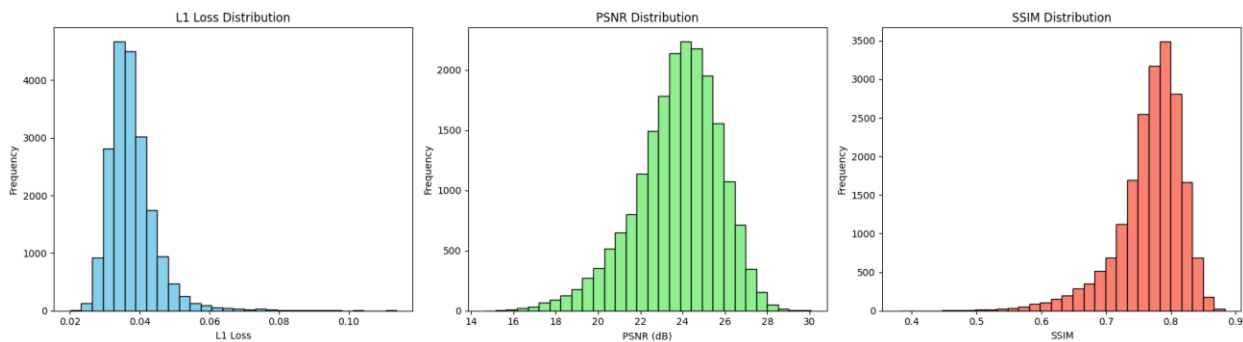


Figure3

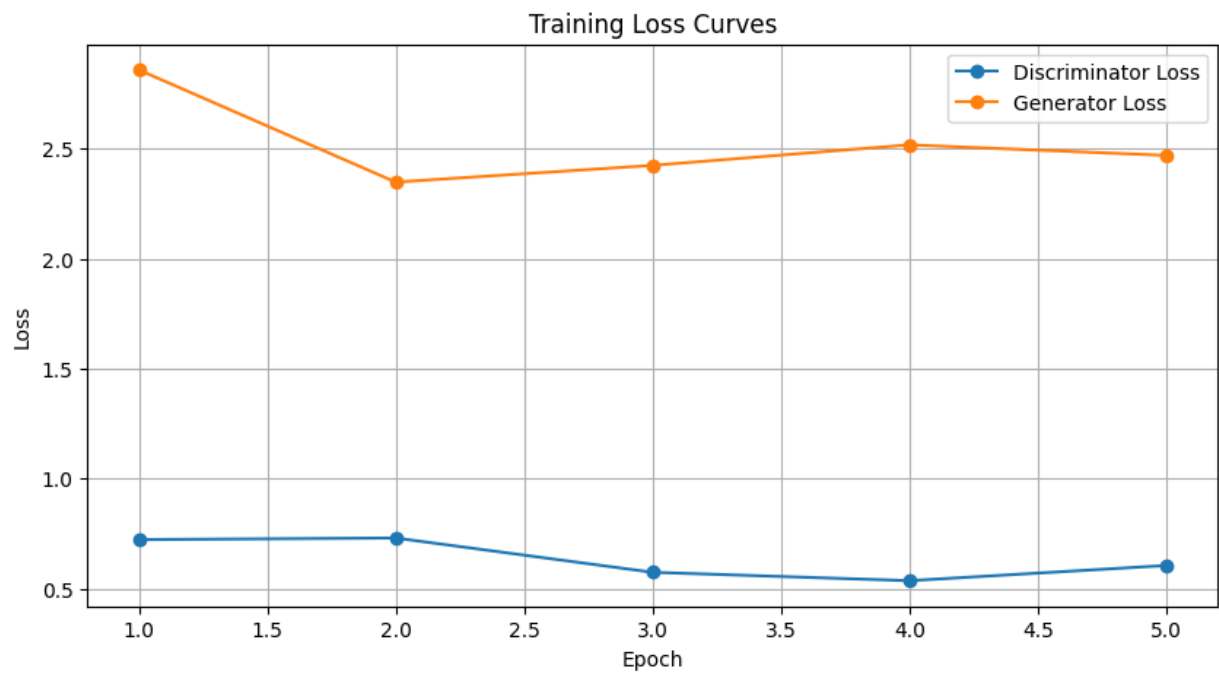


Figure4

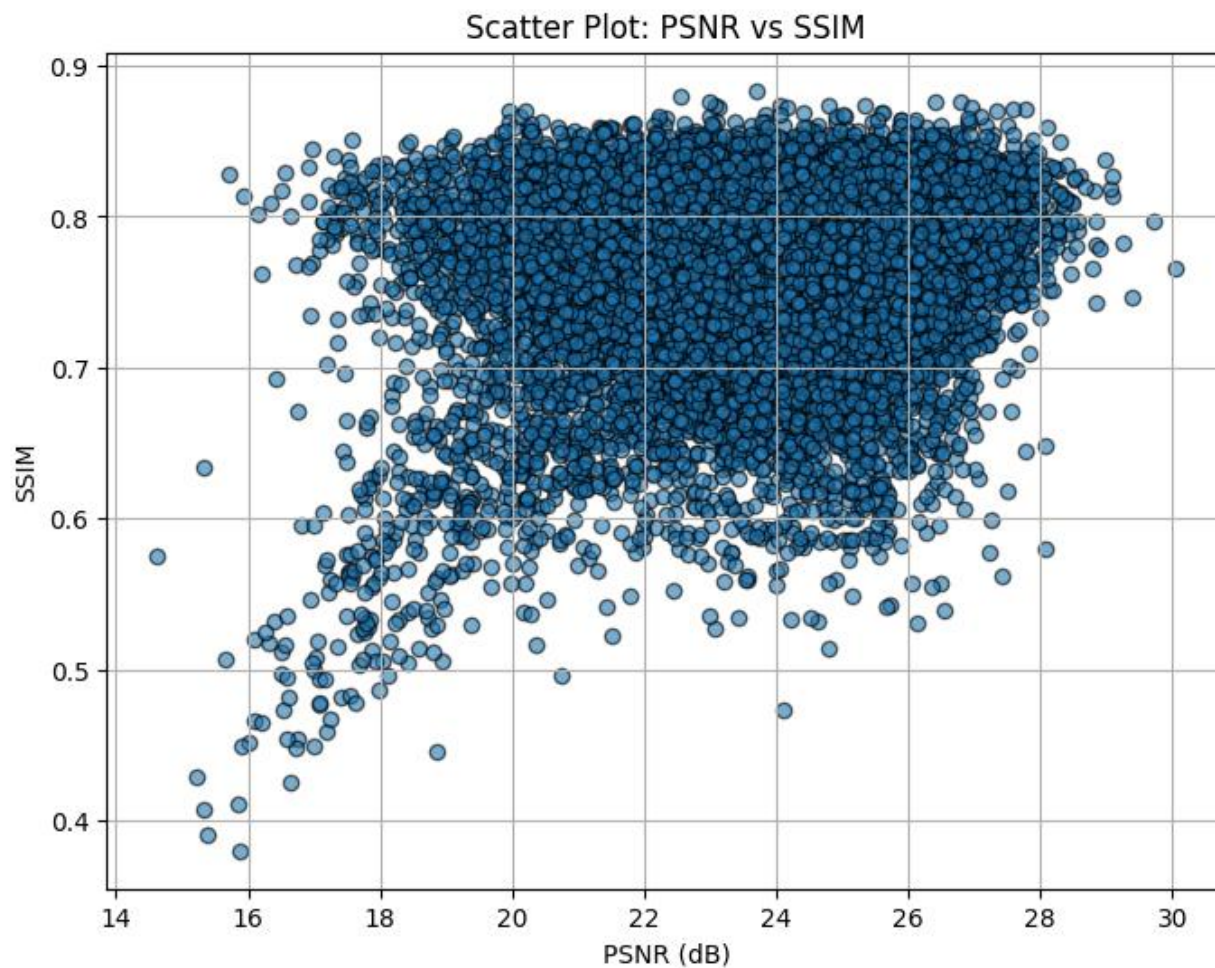


Figure5

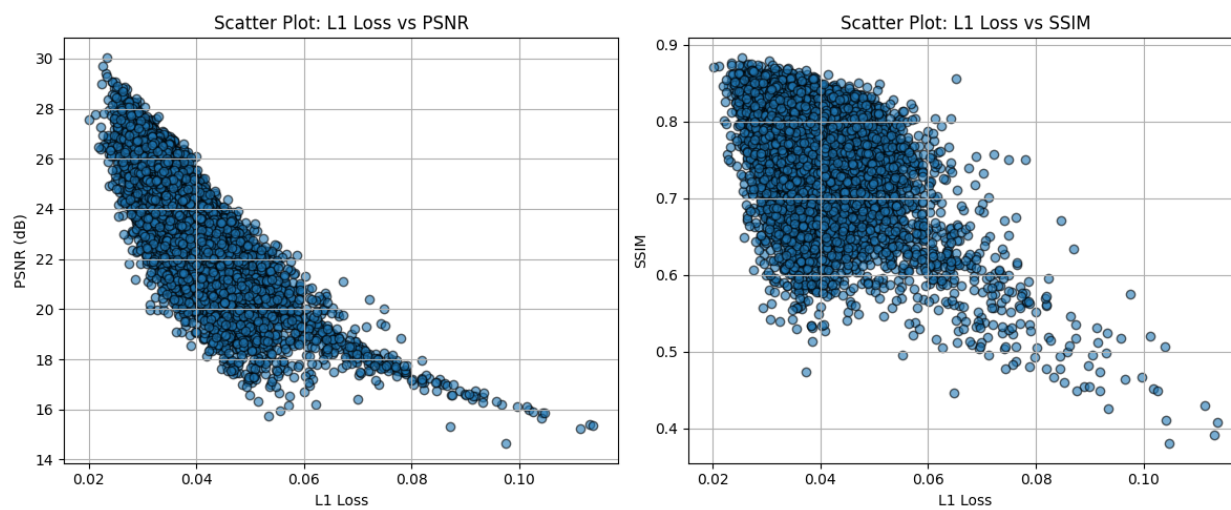


Figure6

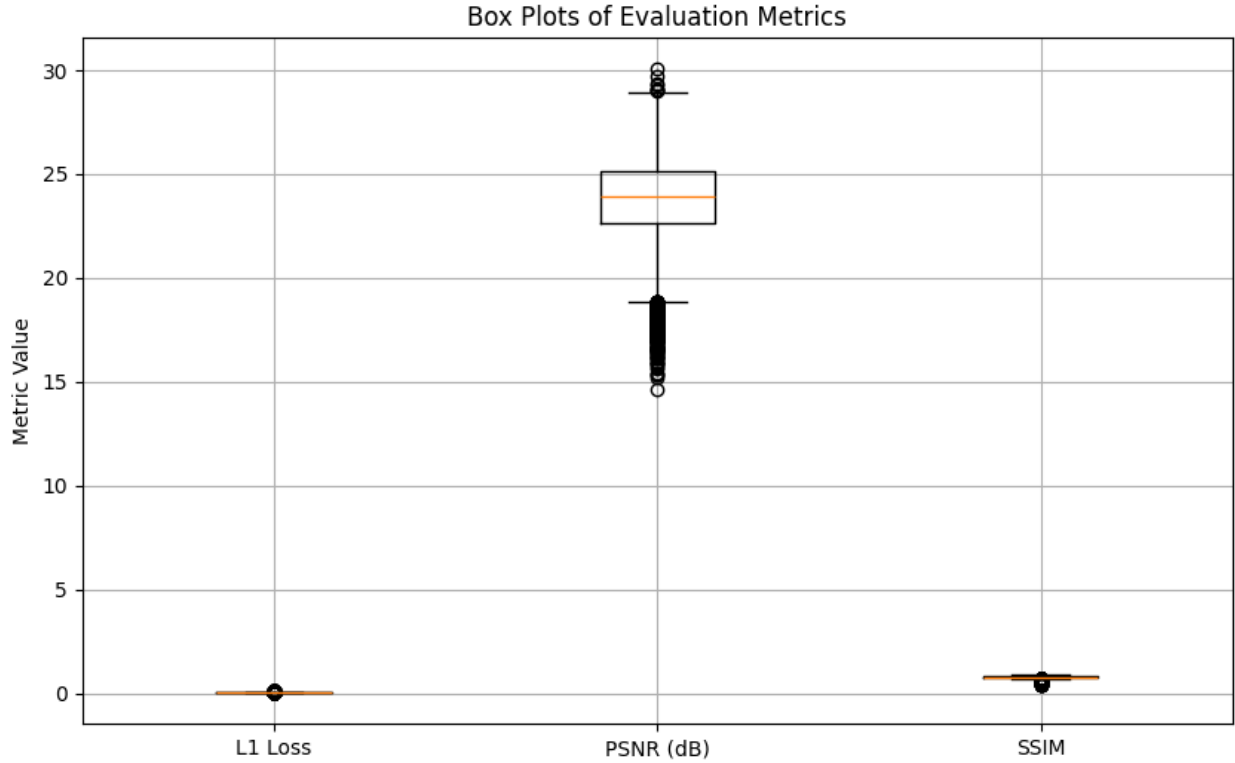


Figure7

We conducted further quantitative analysis to gain more insight into the performance of our hybrid Transformer-GAN model. Figure 3 depicts the distributions of evaluation metrics across our test set, including L1 Loss, PSNR, and SSIM. The plots of distribution reveal that the majority of images possess low L1 loss values at around 0.0377, uniform PSNR values centered predominantly at 23.75 dB, and high structural similarity with the majority of SSIM values being around 0.7674. These histograms verify our model's consistently good reconstruction performance for different samples. In Figure 4, we present training loss curves over five epochs with the progress of the generator and discriminator monitored. The curves are demonstrating stable and convergent behavior with the discriminative loss decreasing progressively and the generator loss stabilizing after oscillation, signaling an optimally balanced adversarial training process. For more insight into metric correlations, we had plotted scatter plots that indicated correlations between different metrics. Figure 5 is a scatter plot of PSNR vs. SSIM, which indicates a positive correlation between the two variables, meaning that images with higher PSNR values have higher structural similarity. Figure 6 presents scatter plots of L1 Loss vs. PSNR and SSIM separately, which clearly demonstrate inverse correlations since lower L1 losses are highly correlated with higher PSNR and SSIM, which verifies the correspondence between pixel-wise accuracy and perceptual quality. Lastly, Figure 7 provides box plots summarizing the distributions of L1 Loss, PSNR, and SSIM, clearly displaying median values, interquartile ranges, and potential outliers. These graphical summaries provide a concise view of the variability and stability of the metrics achieved on the entire test dataset.

In summary, these detailed analyses reinforce our model's efficacy in generating high-quality image inpainting results, tested both visually and quantitatively.

Visual Inspection of Test Samples:

We performed a qualitative visual inspection on several test set samples to directly assess the inpainting capability of the Transformer-GAN hybrid model. Figure 8 contains four sample examples, each including the masked input image, the inpainted output created by our model, the original unmasked image, and the absolute pixel-wise difference between the reconstructed image and the original image. The visualization shows that our model effectively recovered visually coherent content in the masked areas, preserving coherence in facial structures, textures, and colors. Discrepancies between original and reconstructed images indicate where the model struggled, e.g., with subtle color differences or small artifacts near boundaries. However, these qualitative results validate our model's strong capability to produce realistic and accurate inpainting, supporting the quantitative results achieved previously.



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TO DO 0

Create

IN PROGRESS 2

Study existing image inpainting techniques (DeepFill v2, CoMod-GAN, etc.)

☒ GI-3

Research Transformer-based inpainting models

☒ GI-4

DONE 1

Define project objectives and research problem

☒ GI-2

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TO DO 2

Create visuals, graphs, and result analysis

GI-21 SJ

Review project feedback and final refinements

GI-23 TM

+ Create

IN PROGRESS 5

Test model performance on different datasets

GI-16 TM

Compare results with DeepFill v2, LaMa, and CoMod-GAN

GI-17 SJ

Fine-tune hyperparameters for better results

DONE 10

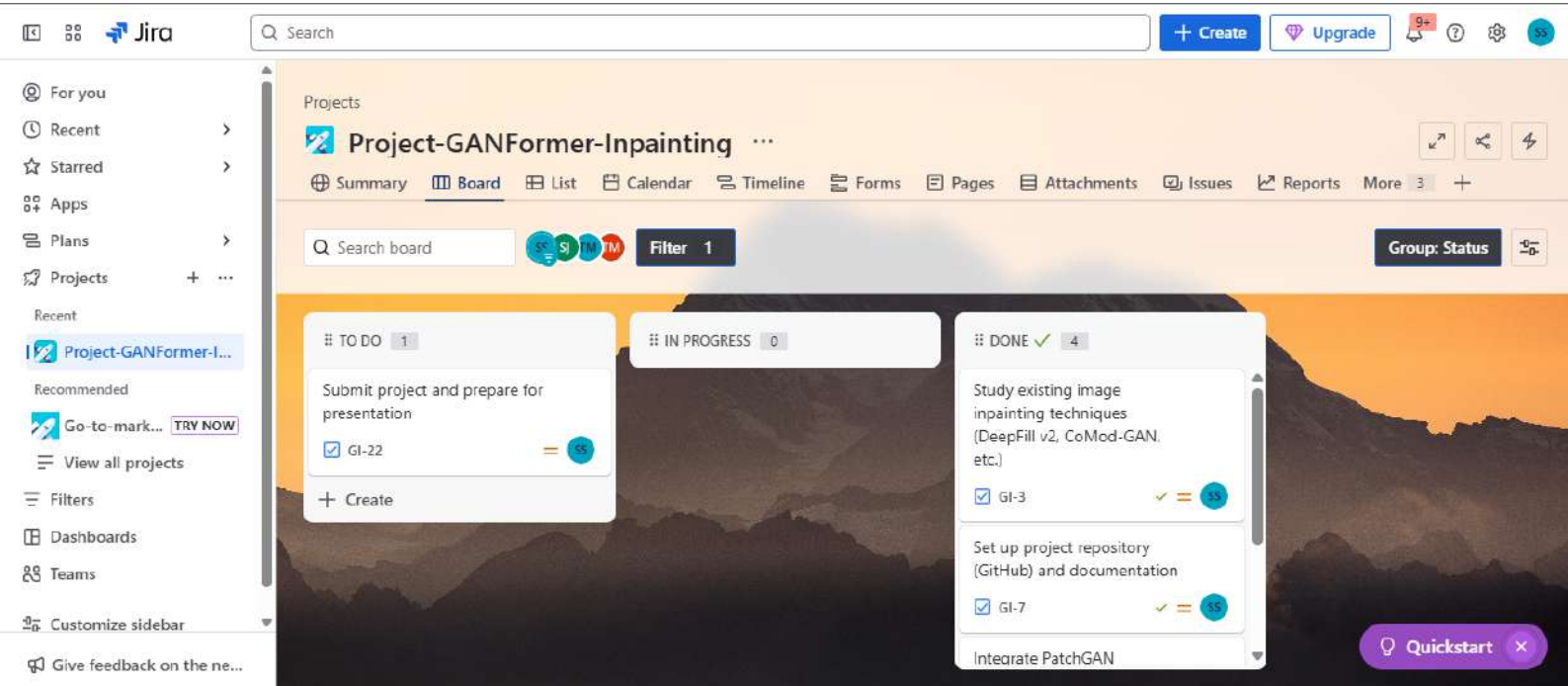
Define project objectives and research problem

GI-2 TM

Research Transformer-based inpainting models

GI-4 SJ

Finalize project methodology and architecture



Projects

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TO DO 2

Submit project and prepare for presentation

☒ GI-22

= 35

Review project feedback and final refinements

☒ GI-23

= TM

+ Create

IN PROGRESS 3

Test model performance on different datasets

☒ GI-16

= TM

Fine-tune hyperparameters for better results

☒ GI-18

= TM

Generate sample inpainted images and analyze outputs

DONE ✓ 8

Define project objectives and research problem

☒ GI-2

✓ = TM

Study existing image inpainting techniques (DeepFill v2, CoMod-GAN, etc.)

☒ GI-3

✓ = 35

Set up project repository

Quickstart

☐	Type	Key	Summary	Status	Comments	Category	+
☐	☒	GI-2	Define project objectives and research problem	DONE	 Add comment		
☐	☒	GI-3	Study existing image inpainting techniques (DeepFill v2, Co...	DONE	 Add comment		
☐	☒	GI-4	Research Transformer-based inpainting models	DONE	 Add comment		
☐	☒	GI-5	Finalize project methodology and architecture	DONE	 Add comment		
☐	☒	GI-6	Identify required datasets and preprocess them	DONE	 Add comment		

☐	Type	Key	Summary	Status	Comments	Category	+
☐	☒	GI-7	Set up project repository (GitHub) and documentation	DONE	 Add comment		
☐	☒	GI-8	Develop Transformer-based encoder	DONE	 Add comment		
☐	☒	GI-9	Implement GAN-based decoder	DONE	 Add comment		
☐	☒	GI-10	Integrate PatchGAN discriminator	DONE	 Add comment		
☐	☒	GI-11	Implement loss functions (adversarial loss, SSIM, perceptual l...	DONE	 Add comment		

☐	Type	Key	Summary	Status	Comments	Category	+
☐	☒	GI-12	Set up the self-supervised pretraining pipeline	DONE	 Add comment		
☐	☒	GI-13	Fine-tune model using inpainting datasets	DONE	 Add comment		
☐	☒	GI-14	Implement evaluation metrics (PSNR, SSIM, FID, LPIPS)	DONE	 Add comment		
 ☐	☒ +	GI-15	Train and optimize the model	DONE	 Add comment		
☐	☒	GI-16	Test model performance on different datasets	IN PROGRESS	 Add comment		

☐	Type	Key	Summary	Status	Comments	Category	+
☐	☒	GI-17	Compare results with DeepFill v2, LaMa, and CoMod-GAN	IN PROGRESS	 Add comment		
☐	☒	GI-18	Fine-tune hyperparameters for better results	IN PROGRESS	 Add comment		
☐	☒	GI-19	Generate sample inpainted images and analyze outputs	IN PROGRESS	 Add comment		
☐	☒	GI-20	Prepare the final report and documentation	IN PROGRESS	 Add comment		
☐	☒	GI-21	Create visuals, graphs, and result analysis	TO DO	 Add comment		

☐	Type	Key	Summary	Status	Comments	Category	+
☐	☒	GI-19	Generate sample inpainted images and analyze outputs	IN PROGRESS	 Add comment		
☐	☒	GI-20	Prepare the final report and documentation	IN PROGRESS	 Add comment		
 ☐	☒ +	GI-21	Create visuals, graphs, and result analysis	TO DO	 Add comment		
☐	☒	GI-22	Submit project and prepare for presentation	TO DO	 Add comment		
☐	☒	GI-23	Review project feedback and final refinements	TO DO	 Add comment		